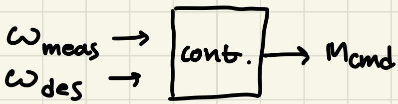
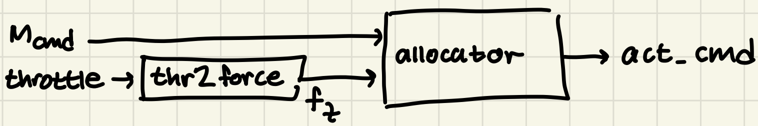
loop

state = bicopter.step(act_cmd)
ω_{body} = state[~]
act_cmd = fsw.get_act_cmd(throttle_frac, ω_{des}, ω_{meas})

fsw

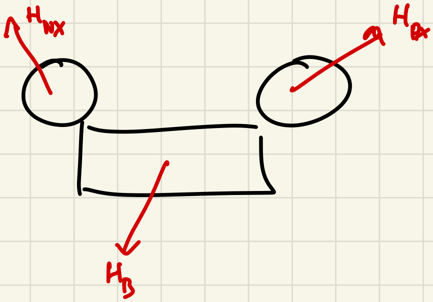
↳ controller (object) needs prior data
↳ allocator (function) ~need prior data
  ↳ vehicle geometry → config["geometric_properties"]
  ↳ throttle2force_lookup





fsw.get_act_cmd(ω_{des}, ω_{meas}, throttle_frac)
  M_{cmd} = self.controller(ω_{des}, ω_{meas})
  act_cmd = self.allocator(M_{cmd}, f_z, vehicle_geometry)

Crabsim V3



take frame derivative

$$H_{PX} = \hat{T} J_{rotor} \omega_{motor}$$

$$\hat{T} = \begin{bmatrix} 0 \\ S\theta \\ C\theta \end{bmatrix}$$

$$M_{PX} + M_{NX} + M_{ext} = M_{NET}$$

$$T = I\alpha$$

$$I^{-1} M_{NET} = \alpha \rightarrow \frac{1}{s} \rightarrow \omega$$

$$H = h_{rotor} \cdot \hat{T}$$

$$\dot{H}_I = \frac{d}{dt}(H)_B + H \times {}^{I}\omega^{B}$$

$$\left(\dot{h}_{rotor}\hat{T} + h_{rotor}\dot{\hat{T}}\right) + H \times {}^{I}\omega^{B} = M_{rotor}$$

200,000 deg/s

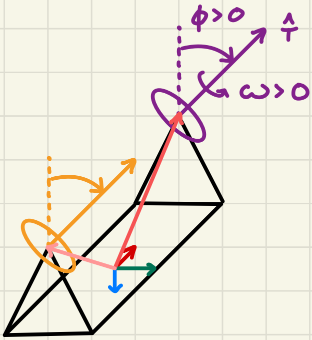
100-200 deg/s

$$H = J_r \omega_r \hat{T} + \cancel{J_t \dot{\theta}_s \hat{S}}$$

~ 0.001% H of rotor
∴ fuck it

$$\dot{x} = Ax + Bu$$

$$y = x$$



$$H_{rotor} = \hat{T}\omega j$$

$$\dot{H}_{rotor} = \frac{d}{dt}_B\left(\hat{T}\omega j\right) + \hat{T}\omega j \times {}^{I}\omega^{B}$$

$$\downarrow$$

$$\dot{\hat{T}}\omega j + \hat{T}\dot{\omega} j + \hat{T}\omega j \times {}^{I}\omega^{B}$$

$$M_{\substack{on\\rotor}} = j\left(\dot{\hat{T}}\omega + \hat{T}\dot{\omega} + \hat{T}\omega \times {}^{I}\omega^{B}\right)$$

$$\therefore M_{on\ body} = -j\left(\dot{\hat{T}}\omega + \hat{T}\dot{\omega} + \hat{T}\omega \times {}^{I}\omega^{B}\right)$$

$$\hat{T} = \begin{bmatrix} 0 \\ s\phi \\ -c\phi \end{bmatrix} \qquad \therefore \dot{\hat{T}} = \begin{bmatrix} 0 \\ s\dot{\phi} \\ -c\dot{\phi} \end{bmatrix}$$

Assumptions

ϕ is only affected by the servos

ω is only affected by the motors

$H_{tilt} << H_{rotor}$